

TMDB Movies Dataset — Data Dictionary

File: `movies.parquet` · Authors: Kseniia Tikhonova, Anastasiia German · Data Acquisition project (M1)

1. Overview

The dataset contains feature-film metadata collected from The Movie Database (TMDB) API v3. It covers movies with a primary release date between 1 January 2000 and 31 December 2024 that had at least 50 user votes on TMDB at the time of collection. Each row describes one movie: its basic details, financial figures, audience ratings, genres, keywords, directors and the top-5 billed cast members.

Property	Value
Source	TMDB API v3 — <code>/discover/movie</code> (movie IDs) and <code>/movie/{id}</code> with <code>append_to_response=keywords,credits</code> (details)
Selection criteria	Primary release date 2000-01-01 to 2024-12-31; <code>vote_count</code> ≥ 50; adult and video titles excluded
Language of text fields	English (<code>language=en-US</code>)
Collection date	2026-10-04 (20:57–21:00 UTC)
Granularity	One row per movie
Primary key	<code>tmdb_id</code> (unique, non-null)
Size	23,981 rows × 16 columns
Format	Apache Parquet, <code>zstd</code> compression, no index column
Missing values	No nulls in any column. Unknown values are encoded as 0 (numeric) or as an empty list (list columns) — see Section 4.
Companion file	<code>acquisition_log.json</code> — request parameters, counts, timestamps and failed requests (none)

2. Table structure

The 16 columns fall into five groups: identification (`tmdb_id`, `title`, `release_date`, `original_language`), production and finance (`runtime`, `budget`, `revenue`), audience reception (`vote_average`, `vote_count`, `popularity`), content descriptors (`genres`, `keywords`) and people (`director_ids`, `director_names`, `cast_ids`, `cast_names`). The last six columns are list columns: each cell holds an array of values. Within a row, `director_ids` and `director_names` are aligned position by position, and so are `cast_ids` and `cast_names`.

#	Column	Parquet type	Group	Short description
1	<code>tmdb_id</code>	<code>int64</code>	Identification	TMDB movie identifier (primary key)
2	<code>title</code>	<code>string</code>	Identification	Movie title in English
3	<code>release_date</code>	<code>timestamp[us]</code>	Identification	Primary release date
4	<code>original_language</code>	<code>string</code>	Identification	ISO 639-1 code of the original language
5	<code>runtime</code>	<code>int64</code>	Production	Duration in minutes
6	<code>budget</code>	<code>int64</code>	Finance	Production budget, USD
7	<code>revenue</code>	<code>int64</code>	Finance	Worldwide box-office revenue, USD
8	<code>vote_average</code>	<code>double</code>	Reception	Mean user rating (0–10)
9	<code>vote_count</code>	<code>int64</code>	Reception	Number of user ratings
10	<code>popularity</code>	<code>double</code>	Reception	TMDB popularity score
11	<code>genres</code>	<code>list<string></code>	Content	Genre names
12	<code>keywords</code>	<code>list<string></code>	Content	Keyword tags
13	<code>director_ids</code>	<code>list<int64></code>	People	TMDB person IDs of the directors
14	<code>director_names</code>	<code>list<string></code>	People	Names of the directors
15	<code>cast_ids</code>	<code>list<int64></code>	People	TMDB person IDs of the top-5 billed actors
16	<code>cast_names</code>	<code>list<string></code>	People	Names of the top-5 billed actors

3. Column descriptions

`tmdb_id`

Unique identifier of the movie in TMDB. It is the primary key of the table and can be used to join with other TMDB data or to build the movie URL (https://www.themoviedb.org/movie/{tmdb_id}).

Type	<code>int64</code>
Source field	<code>id</code> from <code>/movie/{id}</code>
Values in dataset	Unique, no nulls. Range 12 – 1,411,773. IDs are assigned by TMDB and carry no meaning on their own.

`title`

Title of the movie as returned in English (en-US). For non-English films this is the English title if TMDB has one, otherwise the original title.

Type	<code>string</code>
Source field	<code>title</code>
Values in dataset	No nulls. 22,903 distinct values — titles are not unique (remakes and different films share names), so <code>tmdb_id</code> must be used as the identifier. Maximum length 127 characters.

release_date

Primary release date of the movie according to TMDB. This is the date used for the yearly filter during collection.

Type	timestamp[us] (date, time part always 00:00)
Source field	release_date (string YYYY-MM-DD, converted to datetime)
Values in dataset	No nulls. Range 2000-01-01 – 2024-12-31.

original_language

Language in which the movie was originally produced, as a two-letter ISO 639-1 code (e.g. en = English, fr = French).

Type	string
Source field	original_language
Values in dataset	No nulls. 85 distinct codes. Most frequent: en (14,501), fr (2,139), es (1,107), ja (969), it (916).

runtime

Duration of the movie in minutes.

Type	int64
Source field	runtime
Values in dataset	Range 0 – 608, median 98. A value of 0 means unknown (51 rows).

budget

Production budget in US dollars, nominal (not adjusted for inflation). Figures are contributed by TMDB users and may be approximate.

Type	int64
Source field	budget
Values in dataset	Range 0 – 658,800,000. A value of 0 means unknown (15,497 rows, about 65%); these rows should be excluded or treated as missing in any financial analysis.

revenue

Worldwide box-office revenue in US dollars, nominal (not adjusted for inflation).

Type	int64
Source field	revenue
Values in dataset	Range 0 – 2,923,706,026. A value of 0 means unknown (14,121 rows, about 59%).

vote_average

Average rating given by TMDB users on a 0–10 scale.

Type	double
Source field	vote_average
Values in dataset	Range 1.6 – 9.9, median 6.35. Snapshot at the collection date; the value changes over time.

vote_count

Number of user ratings behind vote_average. Useful as a reliability weight for the rating.

Type	int64
Source field	vote_count
Values in dataset	Range 50 – 41,367, median 158 (minimum 50 is the selection threshold). Snapshot at the collection date.

popularity

TMDB popularity score, computed by TMDB from recent activity (page views, votes, watchlist additions, etc.). It reflects current interest, not all-time success.

Type	double
Source field	popularity
Values in dataset	Range 0.006 – 200.5, median 2.72. Highly skewed; snapshot at the collection date.

genres

Genres assigned to the movie, in the order returned by TMDB. A movie can have several genres.

Type	list<string>
Source field	genres[].name
Values in dataset	19 possible values: Action, Adventure, Animation, Comedy, Crime, Documentary, Drama, Family, Fantasy, History, Horror, Music, Mystery, Romance, Science Fiction, TV Movie, Thriller, War, Western. 0 – 9 genres per movie (mean 2.38); 10 movies have an empty list.

keywords

Free-form keyword tags describing the plot, setting or themes of the movie (e.g. "fish", "parent child relationship").

Type	list<string>
Source field	keywords.keywords[].name (via append_to_response)
Values in dataset	22,157 distinct keywords. 0 – 98 per movie (mean 6.61); 3,982 movies have an empty list.

director_ids

TMDB person IDs of everyone credited with the job "Director" in the crew. Using IDs avoids ambiguity between people with the same name.

Type	list<int64>
Source field	credits.crew[].id where job == "Director"
Values in dataset	0 – 36 per movie (mean 1.11); 32 movies have an empty list. Same length and order as director_names.

director_names

Names of the directors, aligned with director_ids.

Type	list<string>
Source field	credits.crew[].name where job == "Director"
Values in dataset	Element i is the name of the person with ID director_ids[i].

cast_ids

TMDB person IDs of the first five actors in billing order. Only the top-5 billed cast is kept to limit the amount of personal data.

Type	list<int64>
Source field	credits.cast[].id, sorted by order, first 5
Values in dataset	0 – 5 per movie (mean 4.88); 133 movies have an empty list. Same length and order as cast_names.

cast_names

Names of the top-5 billed actors, aligned with cast_ids.

Type	list<string>
Source field	credits.cast[].name, sorted by order, first 5
Values in dataset	Element i is the name of the person with ID cast_ids[i].

4. Data quality notes

- **Zero means missing.** In runtime, budget and revenue, TMDB stores 0 when the value is unknown. Convert 0 to NaN before computing averages, ratios (e.g. return on investment) or correlations.
- **Empty lists mean missing.** List columns never contain null; a movie without genres, keywords, directors or cast has an empty list.
- **Snapshot values.** vote_average, vote_count and popularity were captured on 2026-10-04 and will differ if the data is re-collected.
- **Nominal money.** Budget and revenue are in US dollars without inflation adjustment and come from user contributions, so they can be imprecise.
- **Completeness.** The discover step reported 23,982 results; after removing one duplicate ID, 23,981 unique movies were requested and all detail requests succeeded (0 failures).